

THEDFORD, NE, USA

WMO: 722211

Lat: 41.964N

Lon: 100.569W

Elev: 888

StdP: 91.10

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 94087

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-20.2	-17.2	-23.6	0.5	-19.1	-20.9	0.6	-16.1	14.5	-4.8	12.9	-5.8	2.7	290	0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.1	36.0	19.9	33.9	20.1	32.3	19.8	23.0	30.2	22.1	29.3	21.4	28.4	5.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.1	17.6	26.4	20.1	16.5	25.4	19.1	15.5	24.7	73.1	30.2	69.5	29.2	66.5	28.3	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.7	11.1	9.6	-26.0	38.8	1.3	2.3	-26.9	40.5	-27.6	41.8	-28.3	43.1	-29.2	44.8		
(5)			-26.2	24.5	1.2	0.9	-27.1	25.1	-27.8	25.6	-28.5	26.1	-29.4	26.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	-2.5	-2.1	4.5	8.9	14.5	20.8	24.2	22.5	18.0	10.1	3.9	-2.6
	DBStd	10.95	6.66	6.97	6.28	5.45	5.08	3.84	3.37	3.26	4.74	5.27	5.88	6.49
	HDD10.0	1668	389	341	188	83	17	0	0	0	2	63	196	390
	HDD18.3	3504	647	573	428	284	139	18	2	6	64	259	434	648
	CDD10.0	1697	0	2	18	50	156	323	441	389	241	66	12	0
	CDD18.3	491	0	0	1	2	20	92	184	137	53	3	0	0
	CDH23.3	5963	0	1	22	67	326	1117	2183	1459	684	96	9	0
	CDH26.7	2651	0	0	4	16	108	464	1102	657	274	24	1	0
Wind	WSAvg	4.2	4.1	4.1	4.5	5.0	4.7	4.1	3.8	3.7	4.0	4.2	4.0	3.9
Precipitation	PrecAvg	559	11	16	27	66	84	95	77	63	48	42	18	11
	PrecMax	743	36	45	75	157	174	208	178	126	124	142	56	64
	PrecMin	252	1	1	2	9	10	8	15	11	9	2	1	1
	PrecStd	113	8	11	17	36	47	42	39	29	29	32	15	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.1	20.0	26.3	29.1	32.6	36.9	39.4	37.2	35.2	29.9	24.4	17.0
		MCWB	6.8	8.7	11.7	13.6	17.3	19.9	19.5	19.6	18.6	15.4	11.3	7.3
	2%	DB	12.6	15.1	22.2	25.1	29.8	33.8	36.9	34.6	32.6	26.0	20.1	12.4
		MCWB	5.3	6.7	10.1	12.4	17.0	20.2	20.1	19.9	18.9	13.7	9.5	5.4
	5%	DB	9.1	11.3	18.8	22.2	27.2	31.3	34.7	32.5	30.1	22.8	16.7	9.2
		MCWB	3.7	4.6	8.9	11.7	15.8	19.5	20.6	20.1	18.0	12.6	8.0	3.9
	10%	DB	6.4	7.7	15.3	19.1	24.4	29.3	32.5	30.7	27.4	19.9	13.3	6.3
		MCWB	2.2	3.1	7.7	10.5	15.1	19.0	20.5	20.0	16.9	11.5	6.7	2.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.6	9.6	13.8	15.5	20.2	23.4	24.5	23.9	22.1	16.9	12.4	8.4
		MCDB	15.4	19.4	21.3	24.3	28.0	30.7	31.6	30.9	29.0	24.8	21.3	16.0
	2%	WB	5.6	7.0	11.6	14.0	18.6	22.3	23.3	22.6	20.6	15.3	10.3	5.8
		MCDB	12.3	14.5	19.1	22.2	26.6	29.6	30.5	29.1	27.9	22.6	18.3	12.1
	5%	WB	4.0	5.0	9.9	12.6	17.2	21.1	22.4	21.7	19.6	13.6	8.6	3.9
		MCDB	8.7	10.6	17.8	20.4	24.6	28.2	29.9	28.1	27.3	20.9	16.7	8.8
	10%	WB	2.5	3.2	8.0	11.1	15.9	20.0	21.6	20.9	18.6	12.1	6.8	2.2
		MCDB	6.0	7.5	15.5	17.6	22.7	27.2	29.1	27.2	25.4	18.7	13.0	6.0
Mean Daily Temperature Range	5% DB	MDBR	12.2	12.3	13.8	13.3	13.3	13.7	14.1	13.9	14.9	14.2	13.8	12.1
		MCDBR	16.1	17.0	20.0	19.0	17.5	16.6	17.0	16.9	17.9	19.4	18.8	15.5
		MCWBR	9.8	9.7	10.1	8.7	7.2	6.0	4.9	5.7	6.4	9.1	9.8	9.5
	5% WB	MCDBR	14.7	16.4	18.0	16.6	14.8	14.5	14.2	14.0	15.0	16.9	18.1	15.2
MCWBR		9.3	9.8	9.7	8.4	7.0	6.5	5.8	5.7	6.5	8.7	9.5	9.6	
Clear-Sky Solar Irradiance	taub	0.259	0.268	0.292	0.326	0.346	0.365	0.384	0.383	0.342	0.307	0.269	0.260	
	taud	2.534	2.510	2.477	2.391	2.422	2.398	2.388	2.373	2.454	2.520	2.548	2.517	
	Ebn at Noon	913	955	962	944	927	907	884	876	893	893	891	879	
	Edh at Noon	69	83	97	114	114	117	117	115	99	81	67	64	
All-Sky Solar Radiation	RadAvg	2.20	3.09	4.18	5.19	6.11	6.99	7.01	6.03	4.88	3.40	2.41	1.87	
	RadStd	0.17	0.18	0.33	0.29	0.45	0.46	0.24	0.28	0.27	0.34	0.16	0.13	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	+1.30	+1.40	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page